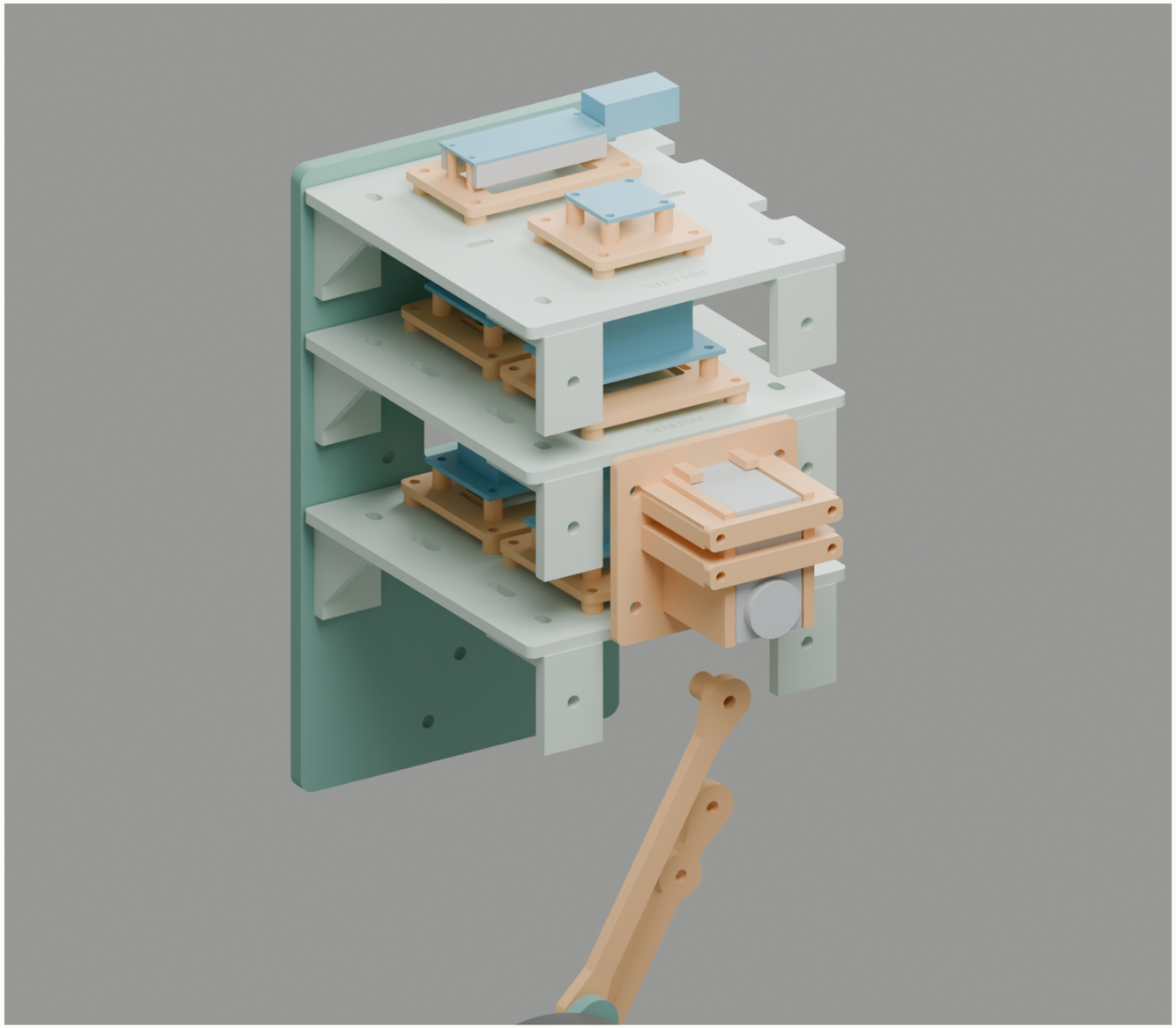
Matched body and removable mounts

Two identical walls / three marked decks / one supported leg / external bench power



CUTAWAY: near wall and fixture omitted. Blue blocks include reserved service space.

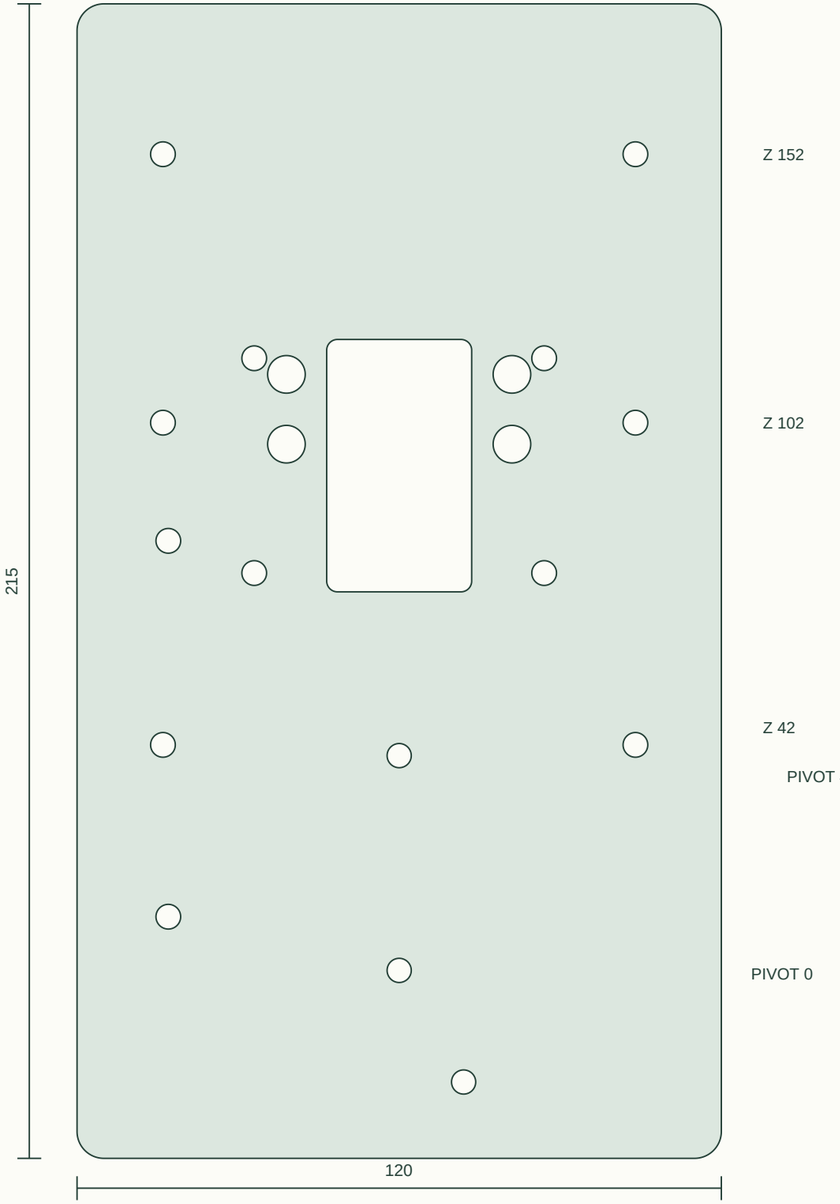
ID	PRINTED PART	QTY
01	Matched body side	2
02	Open deck angle	12
03	Lower power and driver deck	1
04	Middle interface deck	1
05	Upper control and sensor deck	1
06	37D motor carrier with cable anchors	1
07	Lower lock link	1
08	Upper link	1
09	Leg spacer 2 mm	2
10	Leg spacer 12 mm	2
11	Pico 2 plate	1
12	LSM6DSO rigid plate	1
13	ST3215 open case cradle	1
14	ST3215 cap bar	2
15	Driver / power / bus interface cassette	4
16	Pico M2 standoff	4
17	Board M3 standoff	16
18	IMU M3 standoff	4
19	Hole and case width coupon	1
20	Fixture clearance spacer	4
21	Side joint sample	1
22	Deck joint sample	1
23	Module clearance spacer	24

Replace the walls, all decks and all frame angles.
Reuse checked R03 links and leg spacers.
The old motor carrier remains usable if its fit agrees.
Parts 19, 21 and 22 are test pieces.
The cradle is a servo mount. The leg drive remains open.

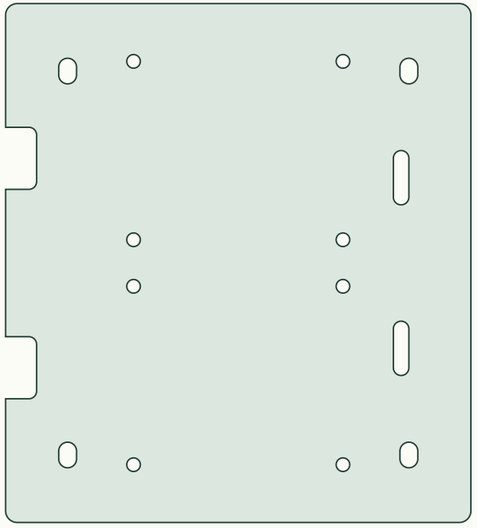
Body and deck interfaces

Contour sections from exported STL / dimensions govern / wall 0.9:1 / decks 0.65:1

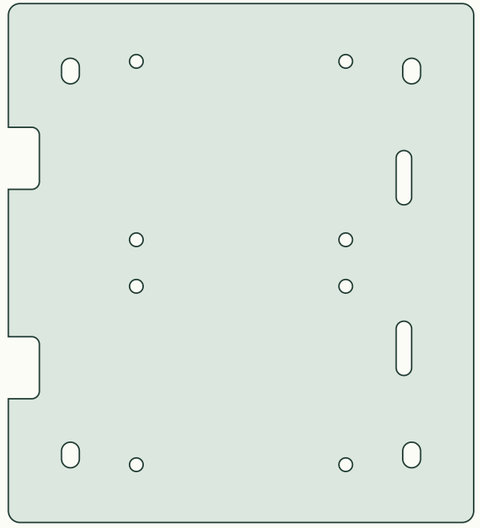
01 / TWO IDENTICAL WALLS



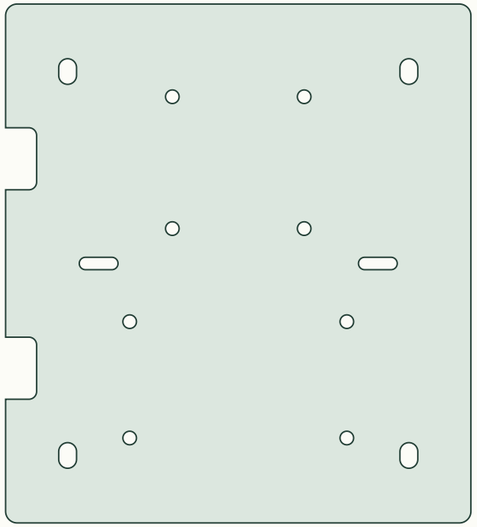
03 / POWER



04 / INTERFACES



05 / CONTROLS



FRAME DATUMS

Near wall: depth 0 to 6
Far wall: depth -141 to -135
Deck center: depth -67.5
Decks: 120 x 133.8 x 4
Side clearance: 0.6 each side
Angle X: -44 and +44
Deck slot depth: -18 and -117
Side bolt Z: deck underside -18

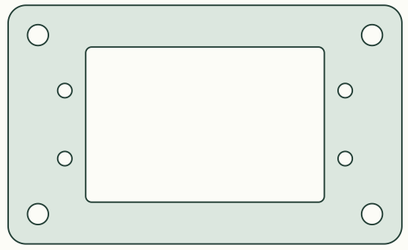
Deck undersides: 60 / 120 / 170
Keep all frame bolts loose until square.

Read README.md. Physical fit, strength and powered qualification remain open.

Electronics and servo attachment

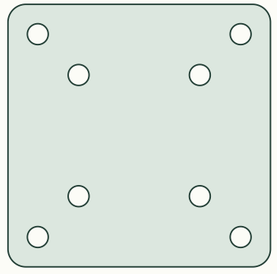
Actual PCB hole patterns / all dimensions in mm / contours at 1:1 unless stated

11 / PICO 2



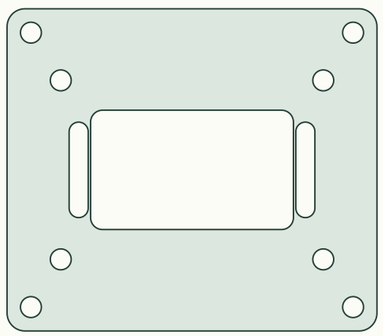
4 x M2, 47 x 11.4 centers
10 high, 3.2 OD standoffs
USB and headers remain accessible.
Use 4 mm feet below this plate.

12 / LSM6DSO



4 x M3, 20.32 square centers
8 mm insulating standoffs
Keep the sensor mount rigid.
Record its axis directions.

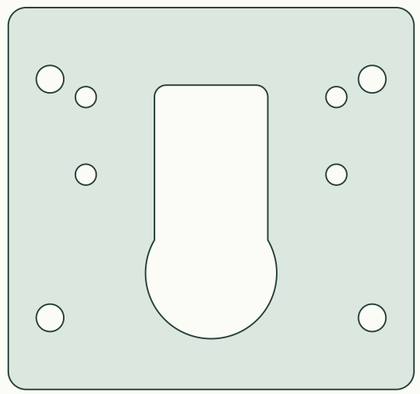
15 / FOUR INTERFACE CASSETTES



Carrier feet: 54 x 46 centers
Board supports: 44 x 30 centers
Board blank: 52 x 38 x 1.6
4 mm feet + 8 mm board standoffs
DRV8874: soldered headers or sockets
Do not drill the driver PCB.
Other board models remain unselected.

Module nuts sit in the 4 mm gap above each deck.
Fit board screws before the module-to-deck screws.
Check actual headers, terminals, cable loops and tool access.

13 / ST3215 CASE CRADLE



Wall bolts: M4, 54 x 40 centers
Case pocket: 25.72 x 46.22
Two cap bars, four M3 x 45 bolts
Rear captive nuts and wall access holes
1 mm resilient pads under bars
Check both output faces and connector.
No powered leg transmission is supplied.

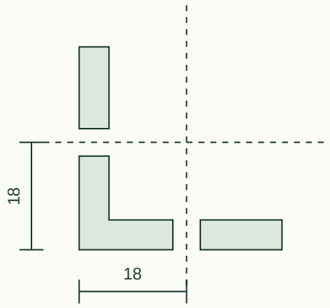


14 / CAP BAR / PRINT TWO

First fit and assembly checks

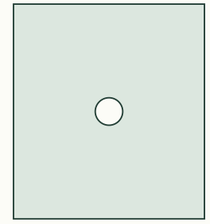
Print one corner before full walls / no print or physical qualification was completed in this session

02 / ANGLE CROSS SECTION / 1:1

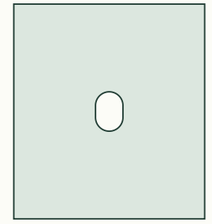


M4 x 16 for each frame connection
Plain 3.2 mm nuts; two 0.5 mm washers per bolt
12 mm OD socket passes the open cavity.
Longer bolts need a new interference check.

21 / SIDE

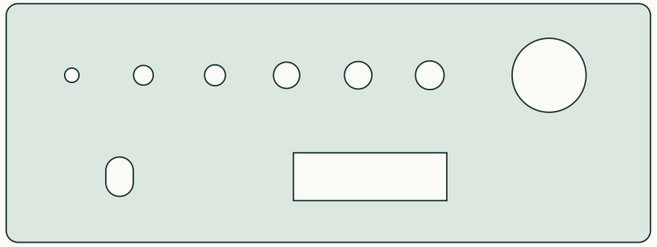


22 / DECK



Print these two samples
and one angle first.
Check both fasteners
with the actual tool.

19 / FIRST FIT COUPON / 1:1



Hole row, left to right:
2.4 / 3.3 / 3.5 / 4.4 / 4.6 / 4.8 / 12.4
Lower slot: 6.6 x 4.6
Lower rectangle: 25.72 wide servo case trial

ASSEMBLY	FASTENER START POINT	CHECK
Frame angles	24 x M4 x 16	Both nuts and the tool fit together
Modules to decks	24 x M3 x 16	4 mm feet leave room for board nuts
Interface boards / IMU	20 x M3 x 16	8 mm insulating standoffs
Pico 2	4 x M2 x 18	10 mm standoffs; inspect header clearance
Motor carrier	6 x M3 x 8	Measure 2 to 3 mm engagement; never exceed 3
Servo bars	4 x M3 x 45	Use pads; do not distort the case

BUILD ORDER
Corner trial, then wall/deck trial.
Matched frame, then electronics.
Independent fixture and catch strap.
Passive leg and neutral lock.
Motor and harness clearance.
Servo arrival and case fit.

Read README.md. Physical fit, strength and powered qualification remain open.